

Are Fish More Stressed in Connected Freshwaters?

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1 Introduction

This project is an investigation into the relationships between fish species richness in UK freshwaters and other types of variables which help determine fish species richness.

The following relationships will be explored:

- The stress put on the fish species richness in fresh waters by the surrounding land cover of the freshwaters (1).
- The relationship between fish species richness in fresh waters and connectivity to other nearby freshwaters (2).
- How the relationship between the stressor variables and species richness is affected depending on how connected the given freshwater is to other nearby freshwaters (3).

The data which has been provided is split into three variable types: connectivity, stressor and typology variables and the response fish species richness. Connectivity variables indicate how connected a freshwater is to another freshwater. There are only two stressor variables in the dataset; the percentage of land covered by urban land in lake catchment and agricultural land in lake catchment.

Typology variables describe the attributes of the freshwater and surrounding catchment, these variables are the control variables. This means that they are used to reduce the effects of confounding when modelling and this will be discussed further in this section 2.

The project aims to further research how connectivity and stressors impact fish species richness, providing explorations into their relationships and interactions so that informed decisions can be made with regard to fish species richness. The focus will be on exploring the impact of connectivity and stressors and how they interact with their ecosystem and influence biodiversity in the UK (Hydroscape, 2016).

The effects of how stressors and connectivity interact and influence the species richness of fish in UK freshwaters is currently poorly understood (Hydroscape, 2016). A better understanding of changes in stressor levels and connectivity can be used to inform environmental and conservation agencies, and policy makers on how to manage freshwaters and the surrounding area (Hydroscape, 2016). Also, understanding how altering land cover, through the introduction of new stressors, may affect the pressures put on the fish species richness is important when new policies are being made, and additionally understanding how an increase in connectivity is not always beneficial since it could lead to a spread in the effect of stressors, such as invasive species and pollution (Hydroscape, 2016).

The data for this dissertation was provided by the Hydroscape project, which is funded by the UK Natural Environment Research Council (NERC) (Hydroscape, 2016). The data was collected by those involved in the project, sampling from three distinct regions of the United

Kingdom: Greater Glasgow, the Norfolk Broads and the Lake District. Each one of these sites has its own unique characteristics, Greater Glasgow being an urban/semi urban district, the Norfolk Broads typically have freshwaters which are highly connected with each other and the Lake District has more upland freshwaters which see less human interaction (UK Lakes Portal, 2021).

The data collected in the Hydroscape project was combined with national scale datasets. For example, data on rivers, obstacles land cover and more, and once these national scale datasets and sampled data were combined, they were used to create connectivity metrics used in this project. The fish species data was collected using three major UK databases and was combined into one database for the sake of the Hydroscape project, the response variable fish species richness comes from these datasets (UK Lakes Portal, 2021).

2 Exploratory Data Analysis

The dataset contains 33 variables and 1111 observations with 32 explanatory variables, 13 of which are connectivity variables, 17 of which are typology variables and there are two stressor variables.

The response variable in the data is fish species richness (will be referred to as *richness*), it represents the counts of different fish species found in a freshwater. The data was provided in an Excel file which was then converted into a CSV file.

The response variable *richness*, is a count variable, it follows a non-normal distribution as can be seen in Figure 1.

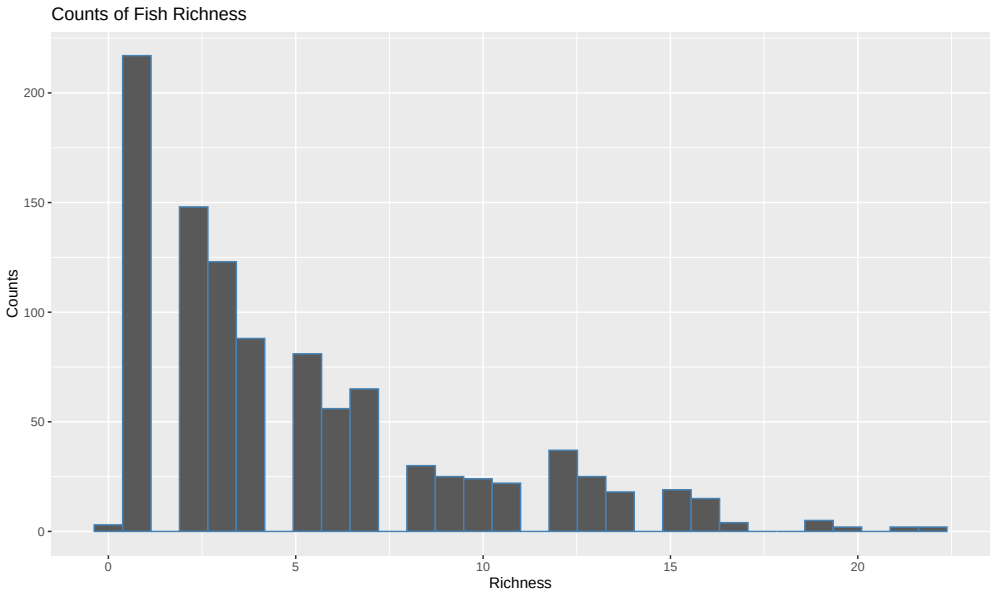


Figure 1: Histogram of Fish Species Richness Counts in UK Freshwaters

Since the response variable follows a non-normal distribution and is count data a Poisson or negative binomial distribution is usually used to model the response (Hilbe, 2014, p. 33). However, before any assumptions about which distribution or model is appropriate the data and the relationships between the variables must be thoroughly explored.

There is no missing data in the dataset. The data indicates that outliers are present, however since it is environmental data there is no reason to remove any outliers as they are all naturally occurring observation in the data (Gla.ac.uk., 2025, Environmental Stats Week 2). For example, the maximum perimeter of a freshwater in the dataset is just over 138.859 km^2 whilst the median perimeter of a freshwater is 1.605 km^2.

Relationships between *richness* and the explanatory were examined. Variables that showed signs of skewness were transformed to ensure that variables selection would not be influenced by extreme observations (Murtaugh, 2009).

As mentioned previously, typology variables are included in order to ensure that the main relationships which are being investigated are not affected by a set of unknown confounding variables and that the relationships of interest between the connectivity, stressor and *richness* are being properly modelled. Confounding occurs when a model is fit and the response variable has a significant relationship with a predictor according to the modelling software, but the predictor in question is not what is actually explaining *richness* (Wood, 2017, p. 36).

Instead, it is another variable which is not in the model or was not even recorded when gathering the data (Wood, 2017, p. 36). To try and avoid this from happening when modelling this data, typology variables were included so that the effects confounding has on the relationships between the stressor and connectivity variables with *richness* is minimised.

Identifying key relationships between the explanatory variables and the response variable is an important step in identifying what type of model is appropriate for the data. Both pre-transformation and post-transformation scatter plots were looked at in order to observe what type of pattern is observed after transformation. Figure 2 shows that the relationships between *richness* and the stressor and typology variables follow a non-linear pattern both pre-transformation and post-transformation. This indicates that a Generalised Additive Model (GAM) should be considered when modelling these relationships (Zuur et al., 2009, p. 37).

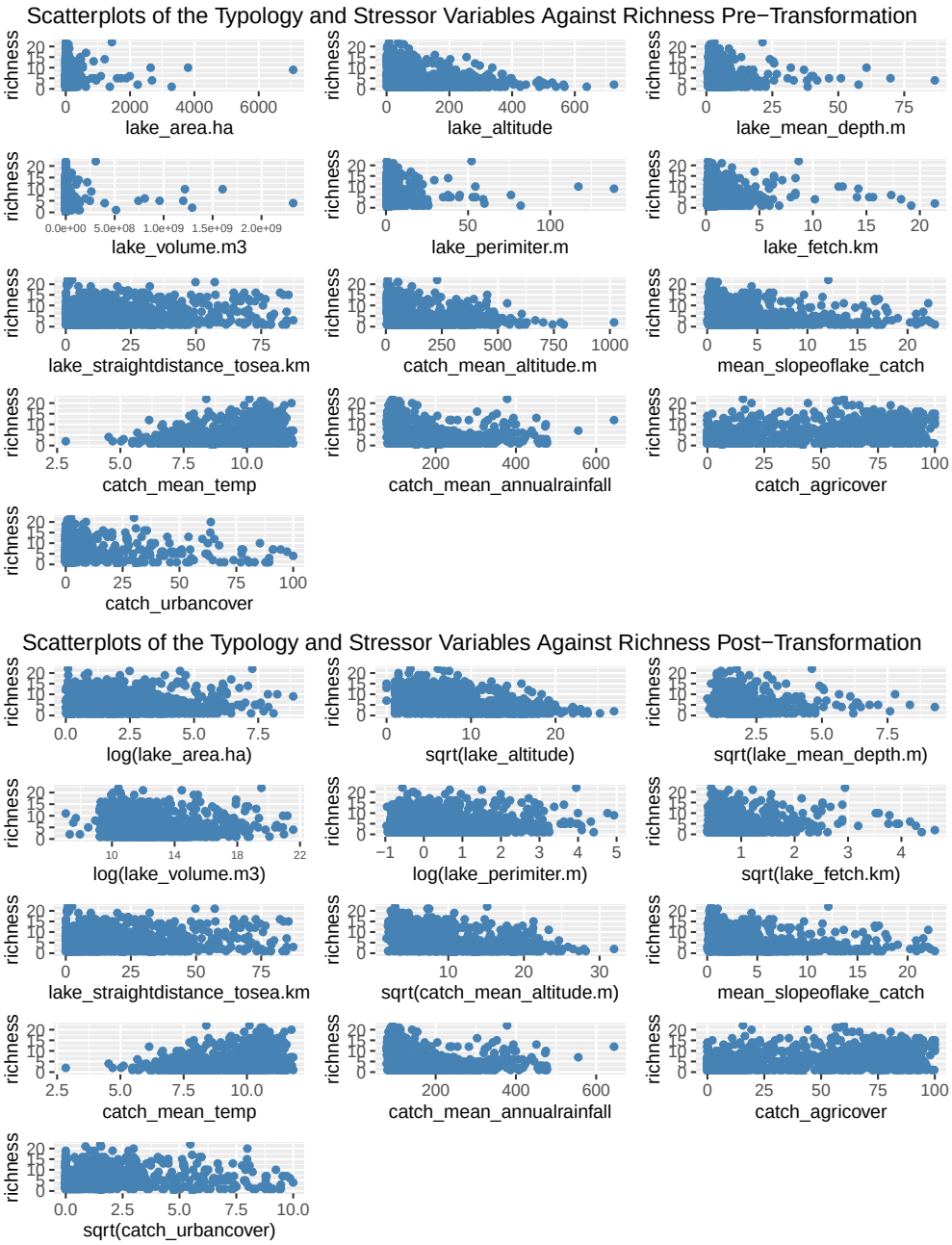


Figure 2: Scatterplot of Typology Variables and Stressor Variables Pre-Transformation and Post-Transformation

Figure 3 shows a similar situation for the relationship between the connectivity variables and *richness*, i.e, a non-linear relationship, again further indicating that a GAM model may be an appropriate model going forward.

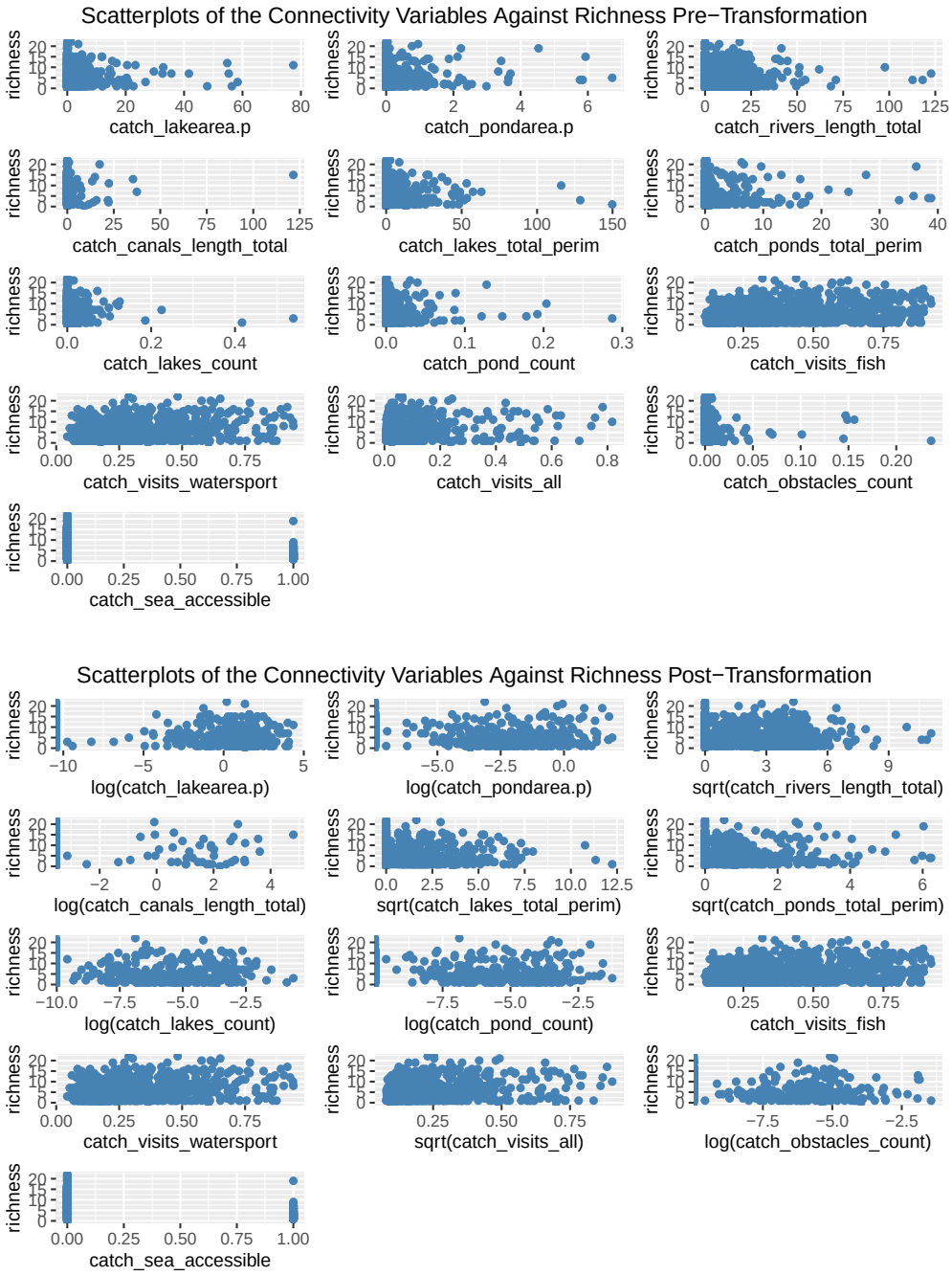


Figure 3: Scatterplot of Connectivity Variables Pre-Transformation and Post-Transformation

To best answer the research questions (1) and (2), two models will be built to assess the relationships between *richness* and the connectivity and stressor variables. It would be difficult to determine the nature of the relationships between the variables of interest and *richness* if both the connectivity and stressor variables were included in the same model.

Pairs plots and correlation matrices were used to identify variables which are highly corre-

lated with one another. Figure 4 shows the correlation matrix for *richness* and the typology and connectivity variables.

It can be seen that many of the typology variables show strong positive linear relationships with other typology variables. When selecting variables to include in a model, especially when working with Generalised Linear Models (GLM) or GAMs, it is important that the explanatory variables are not highly correlated with each other as it will lead to a poorly fitted model (Zuur et al., 2009, p. 63)

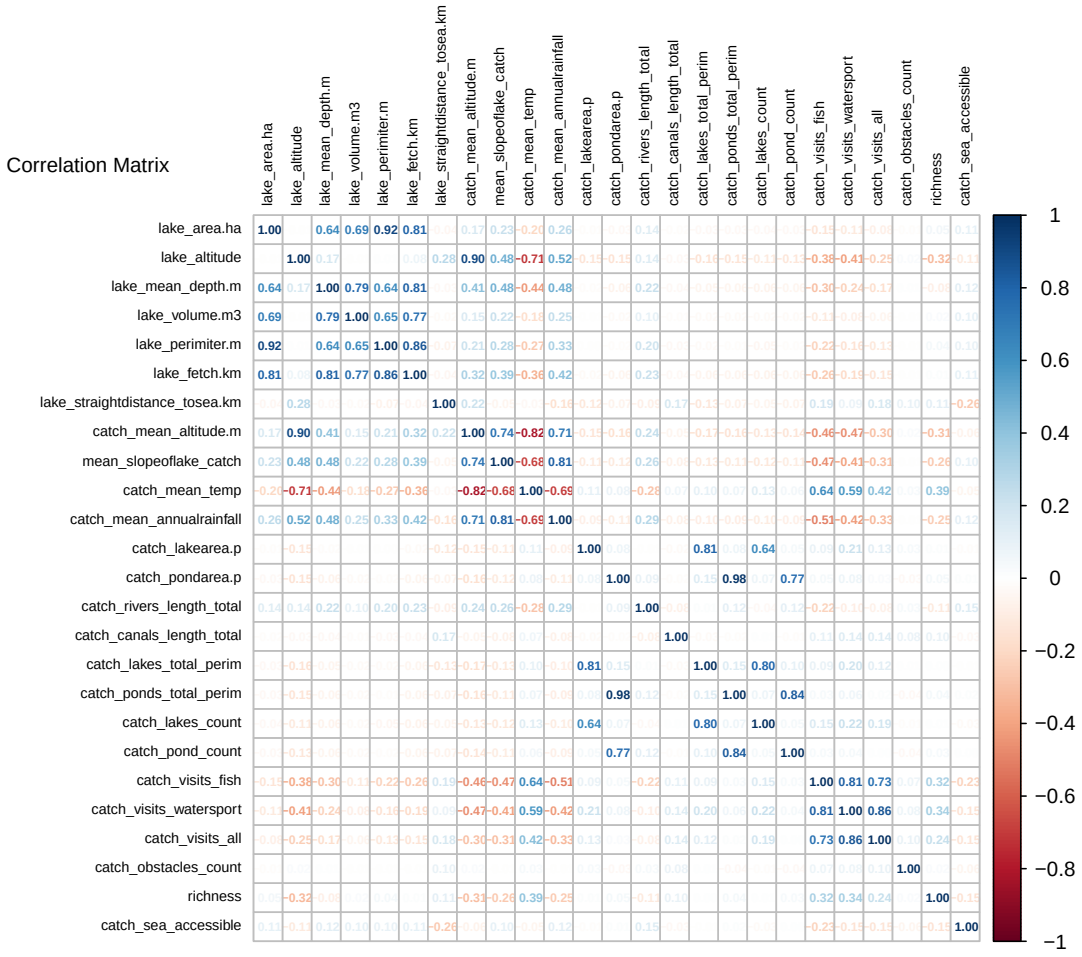


Figure 4: Correlation Matrix of the Typology and Connectivity variables

For example, *lake_area.ha* and *lake_perimiter.m* have a strong positive linear correlation of 0.92, where each of these variables explain lake coverage. It would be unwise to include both

variables when modelling, one of these variables should be chosen over the other. When deciding which of the two to include it is useful to also look at the correlation across all other variables so that multicollinearity can be avoided.

There are many different methods to decide which variables should or should not be included in the model. Figure 4 shows that there is less multicollinearity between the connectivity variables compared to the typology variables.

A common occurrence with the explanatory variables in the dataset will have multiple variables which explain the same thing. For example, only one type of connectivity variable should be chosen to describe the overall presence of other ponds or lakes in the catchment. For example, *catch_lakes_total_perim*, *catch_lakearea.p* and *catch_lakes_count* all are highly correlated with one another however only one should suffice.

3 Description of the Methods/Models

In the E.D.A. it became clear that a GAM may be the appropriate model to handle the apparent non-linear relationships which are present within the data. However, before discussing how a GAM may be used to model the data it is worth mentioning GLMs. GLMs are used in this project to help select variables for the GAMs, they are also investigated as a potential option to help answering the research questions but were eventually ruled out.

3.1 Generalised Linear Models

GLMs allow a non-normal distribution to be modelled for the response variable (Wood, 2017, p. 102), fitting a GLM is similar to fitting a linear model except now a distribution and a link function must be specified (Wood, 2017, p. 101). A GLM may be represented by (3.1)

$$g(\mu_i) = \mathbf{X}_i\boldsymbol{\beta}, \tag{3.1}$$

where $\mu_i \equiv \mathbb{E}(Y_i)$, g is the link function, $\mathbf{X}_i$ is the i-th row of the model matrix and $\boldsymbol{\beta}$ is the unknown parameter vector (Wood, 2017, p. 102). GLMs also make the assumptions that Y_i's are independent.

The reason GLMs are so useful is their ability to model non-normal distributions for the response variable. This is possible for any distribution that falls into the exponential family. A distribution belongs to the exponential family if its probability density function or probability mass function can be written as (3.2) (Wood, 2017, p. 103).

$$f_\theta(y) = \exp\left[\frac{y(\theta) - b(\theta)}{a(\phi)} + c(y, \phi)\right], \tag{3.2}$$

where a, b, and c are arbitrary functions, ϕ is an arbitrary scale parameter and θ is the canonical parameter (Wood, 2017, p. 103).

The exponential family contains some very useful properties, such as the ability to derive the expected value and variance by taking the log-likelihood of the equation, which will be different depending on the given distribution, and then taking the first and second derivatives with respect to θ and rearranging them to find the desired form (Wood, 2017, p. 103).

GLMs do not have a closed form solution for the $\boldsymbol{\beta}$ parameters, instead an estimation is made using an iterative approach. The algorithm is called Iteratively Re-weighted Least Squares (IRLS) (Wood, 2017, p. 107). This algorithm maximises the log likelihood of $\boldsymbol{\beta}$ and uses an iterative method called Newton's method to update the least squares estimate of $\boldsymbol{\beta}$ until convergence has been reached (Wood, 2017, p. 107).

Convergence can be determined in more than one way. Some common approaches are looking at the change in deviance from one iteration to the next or converging when the least squares equation is close to zero. If the algorithm was to diverge at any point in the $\boldsymbol{\beta}^{[k+1]}$ it is possible to move backwards towards $\boldsymbol{\beta}^{[k]}$ (Wood, 2017, p. 107). Distributional results of GLMs are not exact but approximate and rely on large sample limiting results for justification (Wood, 2017, p. 101).

GLMs were used to check which distribution was appropriate to model the data at hand.

3.2 Poisson Distribution

When modelling count data the Poisson distribution is typically looked at first. The Poisson probability distribution function (PDF) represents the count data well (Hilbe, 2014, p. 36), and all other count data models are an extension of basic Poisson models (Hilbe, 2014, p. 36). When modelling with the Poisson distribution certain assumptions have to be met (Hilbe, 2014, p. 36). Of all the Poisson distribution assumptions the one that rarely holds when working with real life data is the assumption that the mean and the variance of the Poisson model are the same, or at least very similar (Hilbe, 2014, p. 36). If this assumption is broken and the variance is greater than the mean then it is a sign that overdispersion is present (Hilbe, 2014, p. 9). Overdispersion is an issue because it leads to underestimation of the standard errors when modelling, which in turn can make it seem like a variable, or variables, in the model are significant predictors of the data when they are not (Hilbe, 2014, p. 82).

There are several checks that may be applied to the Poisson model to assess whether overdispersion is present. The methods used to check for overdispersion in this project were the

calculation of the dispersion statistic and the Score Test. The dispersion statistic is calculated by dividing the Pearson chi-squared statistic, χ^2, by the degrees of freedom. If the dispersion statistic is greater than 1.0 there is evidence of overdispersion (Hilbe, 2014, p. 82).

Before concluding that overdispersion is present, it is wise to check for *apparent* overdispersion. Apparent overdispersion occurs when there is model misspecifications within the Poisson model (Zuur et al., 2009, p. 224) which lead to the characteristics of the data not being properly captured.

The above method was used to check if there was overdispersion present in a Poisson GLM for both a stressor and connectivity model. If there was evidence of overdispersion present in the stressor or connectivity variables then several fixes recommended by Hilbe in *Modeling count data* that can be used to address this. However, after testing there was still evidence of overdispersion as both models had a dispersion statistic greater than 2.0. It was concluded that the Poisson model was not the appropriate choice to model the fish species *richness*.

Since Poisson regression is not appropriate to model the data, the natural choice of distribution is the negative binomial distribution. The negative binomial model has a variance function such that it can model overdispersed Poisson data (Hilbe, 2014, p. 134).

3.3 Negative Binomial Distribution

There are two main parametrisations of the negative binomial distribution NB1 and NB2 (Hilbe, 2014, p. 129). In this project the NB2 parametrisation is used in the MASS package which is used to model GLMs in R.

The negative binomial distribution can be expressed in many different forms. The version below is a common version and comes from Hilbe's *Modeling count data*

$$f(y;\mu,\alpha) = \binom{y_i + \frac{1}{\alpha} - 1}{y_i} \left(\frac{1}{1+\alpha\mu_i}\right)^{\frac{1}{\alpha}} \left(\frac{\alpha\mu_i}{1+\alpha\mu_i}\right)^{y_i} \tag{3.3}$$

Where u is the expected value, and α is the dispersion parameter (Hilbe, 2014, p. 127). The negative binomial model is a two parameter model.

Similar to the Poisson distribution the negative binomial distribution has mean u, but unlike Poisson it has variance $u + \alpha u^2$ (Hilbe, 2014, p. 126). The variance formula for the negative binomial model allows for a wider range of variability compared to the Poisson (Hilbe, 2014, p. 129). Therefore the variance function is able to model overdispersed data (Hilbe, 2014, p. 134).

Since the negative binomial is a two parameter model the question is how can it be modelled

as a GLM? Since there are two parameters rather than one this breaks the assumption that the negative binomial model belongs to the exponential family of distributions (Hilbe, 2014, p. 127). Many software packages use a method to overcome this. They estimate the dispersion parameter as a constant using an algorithm outside of the Iteratively Reweighted Least Squares (IRLS) used in GLMs. It then inserts this dispersion parameter back into the IRLS (Hilbe, 2014, p. 127).

After switching from Poisson to Negative Binomial, the dispersion statistic was checked again for the connectivity and stressor model. The dispersion parameter in both cases was significantly lower and was very close to 1.0 for both models.

3.4 Generalised Additive Models

A GAM is a smoothing model and is a powerful modelling technique. GAMs have the ability to model non-linear relationships between the response variable and multiple explanatory variables (Zuur et al., 2009, p. 36). This is particularly useful for the data in this project as many of the explanatory variables have a non-linear relationship with *richness*. An additive model fits a smoothing curve through the data (Zuur et al., 2009, p. 36), The package *mcgv* which was used to model the GAMs provides several different approaches for smoothing.

GAMs follow the same structure as a GLM. However, now the linear predictors in the model can be modelled as a sum of smoothed explanatory variables (Wood, 2017, p. 161). The structure of a GAM may be represented by,

$$g(\mu_i) = A_i\,\theta + f_1(x_{1i}) + f_2(x_{2i}) + f_3(x_{3i},\, x_{4i}) + \cdots\,, \tag{3.4}$$

where $\mathbf{A_i}$ is the row of the model matrix for parametric model components, $\boldsymbol{\theta}$ is the corresponding parameter vector and $\mu_i \equiv \mathbb{E}(Y_i)$ and $Y_i \sim \mathrm{EF}(\mu_i, \phi)$. Y_i is the response variable, $\mathrm{EF}(\mu_i, \phi)$ is an exponential family distribution with mean u_i and scale parameter ϕ. f_j are the smooth functions of the explanatory variables, x_k, (Wood, 2017, p. 161).

It is extremely useful to be able to include a smooth function of an explanatory variable, or variables, when modelling without being specific about the exact form the function will take (Wood and Augustin, 2002, p. 159). This is done using basis functions and basis functions represent $f(x)$ so that it becomes a linear model and therefore f can be estimated more easily (Wood, 2017, p. 162). The basis for $f(x)$ comprised of basis functions $b_j(x)$ takes the form

$$f(x) = \sum_{j=1}^{k} b_j(x)\,\beta_j \tag{3.5}$$

where β_j are a set of unknown parameters.

There are many ways to optimise and fit a GAM. Cubic regression splines is a method that uses Ordinary Least Squares (OLS) to obtain the fitted values (Zuur et al., 2009, p. 45). The number of knots (smoother connection points) is used to determine the wiggliness of the smoother. The placement of the knots is either through using quartiles or equidistant positions (Zuur et al., 2009, p. 48).

Choosing the number of knots can be problematic because if there is uneven knot spacing this can lead to poor performance (Wood, 2017, p. 167). The fitting of the regression models using this method is heavily dependent on the location of the knots (Wood, 2017, p. 167). Another method called penalised regression splines, which is used in this project, fixes the problems previously mentioned.

Instead of choosing the optimal number of knots as with cubic regression splines, now with penalised regression splines the basis dimension is fixed at a size slightly larger than is assumed to be necessary (Wood, 2017, p. 167). The $\boldsymbol{\beta}$ are found by minimizing the least squares fitting objective but now an extra penalty parameter λ is added to penalise wiggliness of the fitting objective which can be seen in (3.6) (Wood, 2017, p. 167).

$$\|\boldsymbol{Y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 + \lambda \int f''(x)^2 \, dx \tag{3.6}$$

where $\mathbf{Y}$ contains all the observed data in vector format, $\mathbf{X}$ contains all the $b_j s$ and $\boldsymbol{\beta}$ contains all the parameters in vector format (Zuur et al., 2009, p. 49).

The second derivative f'' indicates how smooth a curve is where a large value of f'' means that the smoother f is highly non-linear (Zuur et al., 2009, p. 49). The smoothing parameter λ controls the wiggliness of the smoother, with $\lambda \to \infty$ leading to a straight line estimate for f, whilst $\lambda = 0$ will lead to no penalty at all and the curve massively overfitting the data (Wood, 2017, p. 108).

With the penalised regression method, instead of wanting to optimise the number of knots, the goal now is to select the optimal value of λ, such that, $\hat{f}$ is as close to the true function as possible without being under smoothed or over smoothed (Wood, 2017, p. 169). There are several methods to optimise the selection of λ: Restricted Maximum Likelihood (REML) is the method used in this project. Generalised Cross validation (GCV) is another popular method where cross validation is used to find the optimal value of λ (Zuur et al., 2009, p. 51).

To estimate a smoother when working with a GAM, the smoothing parameter λ will be estimated using whichever criteria has been chosen (in this project REML was used). The process of estimating λ is called outer iteration (Wood, 2017, p. 269). Once the smoothing parameter(s) have been estimated an inner iteration will be used to find the model coefficient estimates (Wood, 2017, p. 269). The GAM is fitted using penalised likelihood maximisation and this is achieved by using Penalised Iterative Least Squares (PIRLS) (Wood, 2017, p. 180). The outer iteration uses Newton's method to ensure reliability which will require

finding the first and second derivative of the model coefficient estimates with respect to λ (Wood, 2017, p. 169). This process of outer and inner iteration is continued until convergence has been achieved and suitable estimates for the GAM have been obtained. This above approach was adopted when modelling the GAMs in this project.

3.5 Framework for Variable Selection

As discussed in Section 2 variable selection is very important when building the model that best represents the data and which can be used to explain the relationships between the variables. When subject matter knowledge is limited the use of data driven techniques to select the variables is a sensible strategy, (Sauerbrei et al., 2007, p. 5) and this was the approach used to create a general framework for variable selection.

The general framework for variable selection used to build the models in this dissertation is as follows:

- Use a correlation matrix to assess correlation between variables, where ± 0.75 was the threshold for highly correlated variables and under ± 0.3 the threshold for lack of correlation. This step was taken so that variables with high correlation were noted and were not used in the same model.

  Note, as discussed in Section 2, transformations were applied in cases of skewness, prior to the next steps

- Backwards selection is performed on a GLM with a negative binomial distribution, where the negative binomial was used to handle the overdispersion. The selection criteria for the backwards selection uses both AIC and BIC.

- Once the GLMs were constructed the variables chosen by backwards selection were used as a basis when building the GAMs

  Some variables which were not chosen by the backwards selection for the GLM but were thought may have an impact when modelling were also selected for the GAM. In this instance subject matter knowledge from literature was used to help in the selection. (Sauerbrei et al., 2007, p. 8)

- Various GAM with different variables were compared. It was also ensured that no variables with strong correlation were added to avoid multicollinearity.

- Each candidate model was assessed based on residual diagnostics, interpretability, and BIC values. In most cases the model with the best BIC value was also the model in which the variables were most interpretable in an environmental context. The final GAM models built whilst containing variables used in the original GLMs were not the same due to the properties of the GAM and its ability to model non-linear relationships which the GLM could not capture.

4 Results/Analysis of the Data

An effective way to examine how the *smooths* of our explanatory variables interact with *richness* is to look at the estimated smoothing curve. In these curves the x-axis shows the values of the explanatory variables and the small vertical lines along the x-axis are rug plots which indicate the values of the predictor along the x-axis. The y-axis shows the contribution of the smoother to the fitted values (Zuur et al., 2009, p. 58). The solid line is the smoother and the dotted lines are the 95% confidence intervals (Zuur et al., 2009, p. 58).

It is worth noting that the typology variables which were included in the models built were always significant and usually exhibited a higher degree of non-linearity then the variables of interest which were examined. An example of this can be seen that was taken from the stressor model can be seen in Figure 5 below.

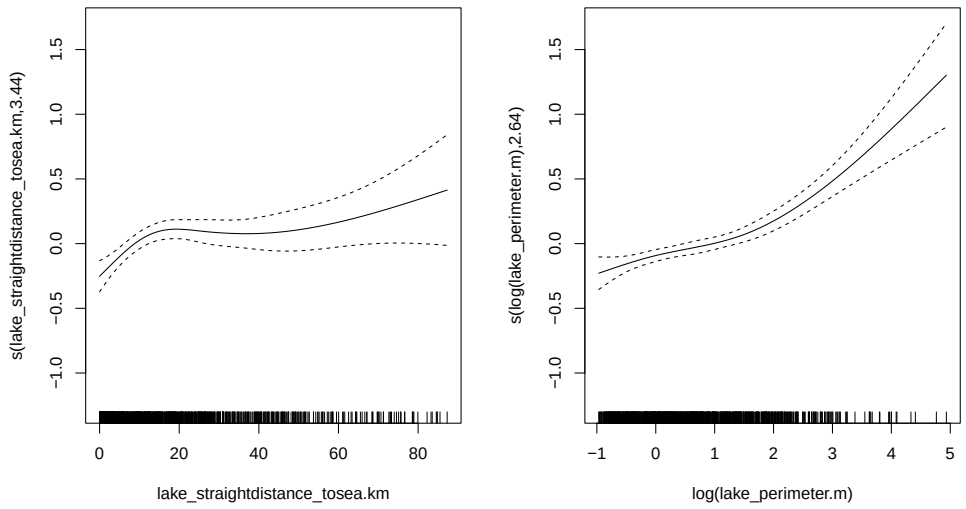


Figure 5: Estimated smoothing curves for *lake_straightdistance_tosea.km* (left) and *lake_perimeter.m* (right).

The GAMs built in this project used cubic regression splines on variables with a single smooth and Gaussian process basis was used to model the longitude and latitude, this basis was chosen in order to help with spatial autocorrelation. As previously mentioned a negative binomial distribution with a log link was used to model the data.

4.1 Research question 1: the affects of stressor variables on *richness*

The stressor model built included the two stressor variables (% urban and agricultural land cover) and several typology variables. The model equation can be seen below

$$\begin{aligned}\log\big(\mathbb{E}[\text{richness}_i]\big) &= f_1(\text{lat}_i, \text{lon}_i) + f_2\big(\sqrt{\text{catch_mean_altitude}_i}\big) + f_3(\text{lake_straightdistance_tosea}_i) \\ &\quad + f_4\big(\log(\text{lake_perimeter}_i)\big) + f_5\big(\sqrt{\text{catch_urbancover}_i}\big) + f_6(\text{catch_agricover}_i) \\ &\quad + \sum_{j=1}^{J-1} \beta_j \, \mathbf{1}\{\text{lake_riverbasindistrict}_i = j\}. \end{aligned} \tag{4.1}$$

As seen in Figure 6, *catch_urbancover* has a non-linear relationship with *richness*. The variable *catch_urbancover* has an Effective Degrees of Freedom (EDF) of 1.24, it is non-linear but not extremely wiggly. As the percentage of urban land cover increases the partial effect on *richness* decreases. The opposite can be said for *catch_agricover*, its EDF value is 1.0 and therefore it has a linear relationship with *richness*. It can be seen that *catch_agricover* has a positive increasing partial effect on *richness*, so, as the percentage of agricultural land increases in a catchment, the counts of different fish species also increases.

When modelling the effects of the two stressor variables only *catch_agricover* had a significant effect on the response variable *richness*, *catch_urbancover* was not significant. The significance of the variables in the model was checked using the p-values provided. All of the typology variables in the stressor model were significant when modelling *richness*.

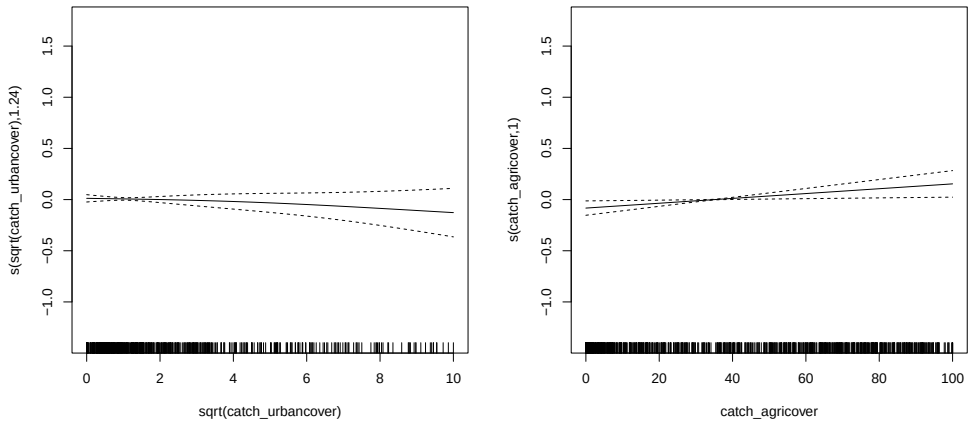


Figure 6: Estimated smoothing curves for *catch_urbancover* (left) and *catch_agricover* (right).

4.2 Research question 2: the relationship between *richness* and connectivity to other nearby freshwaters

The connectivity model was challenging to build due to the nature of the connectivity variables. All of the connectivity variables describing the connectivity of the catchment had a very high proportion of zeros, in fact the variables were made up of 60% of zeros. Except for the connectivity variables describing human visits to and interactions with the catchment and the variable describing the presence of rivers in the catchment.

Therefore it was difficult to find tangible relationships between the connectivity variables and *richness*. Previously mentioned in the project, confounding can lead statisticians to believe that they are modelling significant relationships when they are not. To ensure this was not happening several significant typology variables were included in the connectivity model. In order to ensure that the best model was selected to examine the relationships of interest, and since it was difficult to find significant connectivity variables, other metrics were used to select the best model. The model that was chosen minimised the BIC value, and had the highest amount of deviance explained as well as the highest adjusted R-squared.

The model equation for the connectivity model can be seen in (4.2).

$$\begin{aligned}\log\left(\mathbb{E}[\text{richness}_i]\right) &= f_1(\text{lat}_i, \text{lon}_i) + f_2(\log(\text{lake_perimeter}_i)) \\ &\quad + f_3\Big(\sqrt{\text{lake_altitude}_i}\Big) + f_4(\text{lake_straightdistance_tosea}_i) \\ &\quad + f_5\Big(\sqrt{\text{catch_lakes_total_perim}_i}\Big) + f_6(\text{catch_visits_watersport}_i) \\ &\quad + \sum_{j=1}^{J-1} \gamma_j \, \mathbf{1}\{\text{lake_riverbasindistrict}_i = j\} + \delta \, \mathbf{1}\{\text{catch_sea_accessible}_i = 1\}.\end{aligned} \tag{4.2}$$

As can be seen in Figure 7 both *catch_lakes_total_perim* and *catch_visits_watersport* both have a linear relationship with *richness*. The variable *catch_lakes_total_perim* is the total perimeter of all the other lakes in the catchment divided by the catchment area. In essence, the variable is used to describe the presence of other lakes in the catchment. As *catch_lakes_total_perim* increases the partial effect on *richness* decreases. However, it appears that this is driven by a few outliers that can be seen towards the end of the distribution of *catch_lakes_total_perim*. The *catch_visits_watersport* is the normalised measure of number of visits for watersports purposes in lake catchment, which has an increasing partial effect on *richness*.

Although both of these variables have an effect on *richness*, neither of these variables are significant with p-values of 0.25 and 0.28 respectively. The other connectivity variable in the model is *catch_sea_accessible*. It is a binary variable, an observation is given the value 1 if the freshwater is accessible to fish from the sea and 0 if not. Since it is a categorical variable it

was not smoothed but instead entered into the model as a parametric term. It has a p-value of 0.063, so it has the lowest of all the connectivity variables and the coefficient estimate is -0.15, so this model suggests that if a freshwater is accessible to the sea it may have fewer fish species counts.

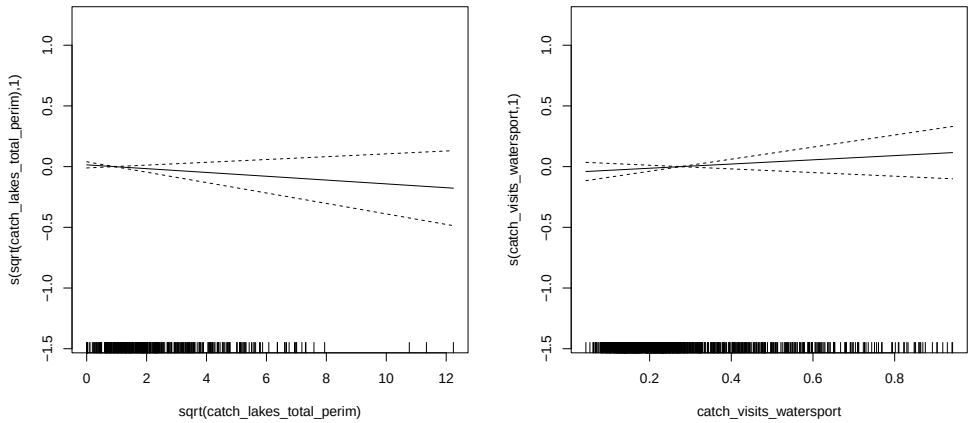


Figure 7: Estimated smoothing curves for *catch_lakes_total_perim* (left) and *catch_visits_watersport* (right).

4.3 Research question 3: how the relationship between stress and *richness* is affected by connectivity

The approach taken to explore the relationship between the stressor variables in the data and *richness* and how they are affected by the connectivity variables is to build a GAM and investigate the interactions between the stressor and connectivity variables.

However, this approach also suffers from a similar issue to that seen in the connectivity model. It is difficult to find tangible interactions between the stressor variables and connectivity variables. Therefore, when building this model, deviance explained, adjusted R-squared and BIC were used as metrics of goodness of fit. However, this time it is worth noting that the lowest BIC model was not selected. This is because BIC penalises model complexity and the interaction terms which are worth looking into would not have been included in the model, and the lowest BIC would favour a model containing only one interaction of interest. Since the question is to assess the relationships between the stressor and connectivity variables and how it affects *richness* a model with slightly worse BIC, similar adjusted R-squared and deviance explained was selected.

The model that was used combined all the variables in the connectivity model bar *catch_visits_watersport* and all the variables stressor model and *catch_mean_altitude.m* was

chosen over *lake_altitude*. This time interaction terms were added to the model to explore the interactions of interest. The interactions that will be explored are

$$f_4(\text{catch_agricover}_i, \text{catch_lakes_total_perim}_i),$$
$$f_5(\text{catch_urbancover}_i, \text{catch_lakes_total_perim}_i),$$
$$f_6\big(\text{catch_agricover}_i \mid \text{catch_sea_accessible}_i = 1\big)$$

Looking at the estimated smooths of the interaction terms in Figure 8 it can be seen for $f_4(\text{catch_agricover}_i, \text{catch_lakes_total_perim}_i)$ that as *catch_agricover* and *catch_lakes_total_perim* increase in the same direction so does *richness* as well. This indicates that the interaction between these two variables has a positive increase on the response.

The oppisite can be said for $f_5(\text{catch_urbancover}_i, \text{catch_lakes_total_perim}_i)$. As *catch_urbancover* increases in the same direction in tandem with
catch_lakes_total_perim a decrease in *richness* can be observed, *richness* is at its highest when *catch_urbancover* and *catch_lakes_total_perim* are at their lowest. However, it is worth noting similar to before, that neither of these interactions is significant with a p-value of 0.35 and 0.33 respectively.

The third and final interaction to be explored is $f_6\big(\text{catch_agricover}_i \mid \text{catch_sea_accessible}_i = 1\big)$. Looking at Figure 9 it can be seen that the as the interaction between *catch_sea_accessible* and *catch_agricover* increases it has a negative partial effect on *richness*.

This interaction is also non-linear with an EDF of 1.32, the confidence bands are large so there is some uncertainty about the effect. This was the only significant interaction term having a p-value of 0.04. The residual diagnostics of the interaction model is excluded from the report as it is very similar to the residual diagnostics of the other two variables, it can be found in the appendix.

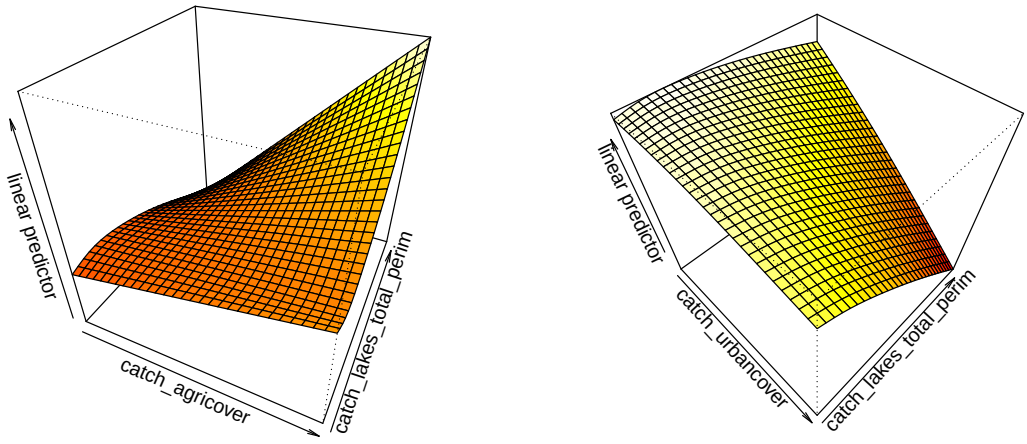


Figure 8: 3D estimated smoothing curves for the interaction of *catch_agricover* and *catch_lakes_total_perim* (left), and for the interaction of *catch_urbancover* and *catch_lakes_total_perim* (right).

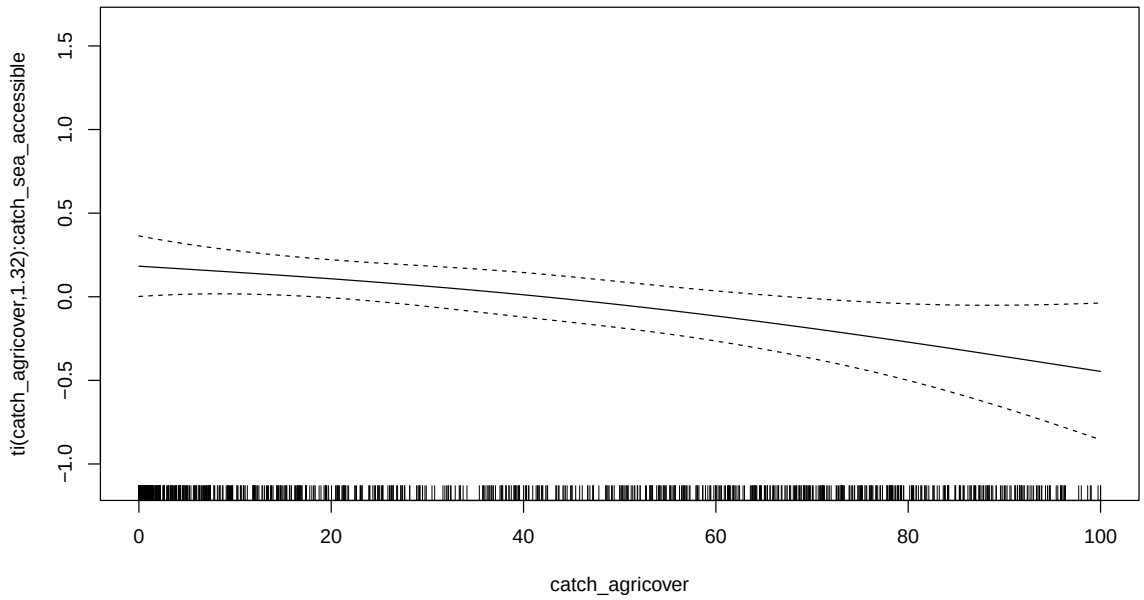
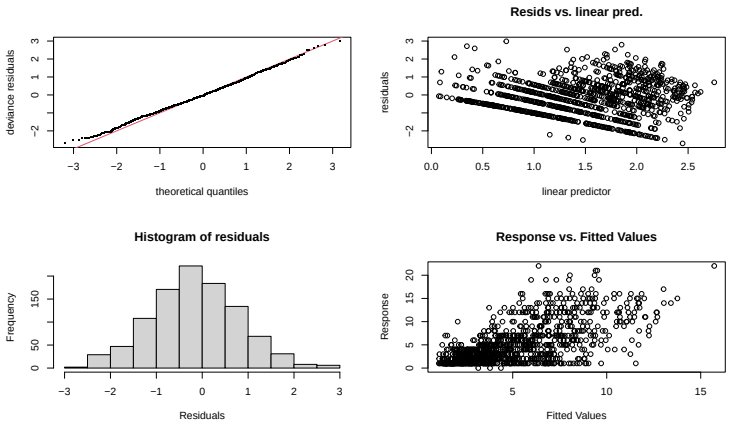
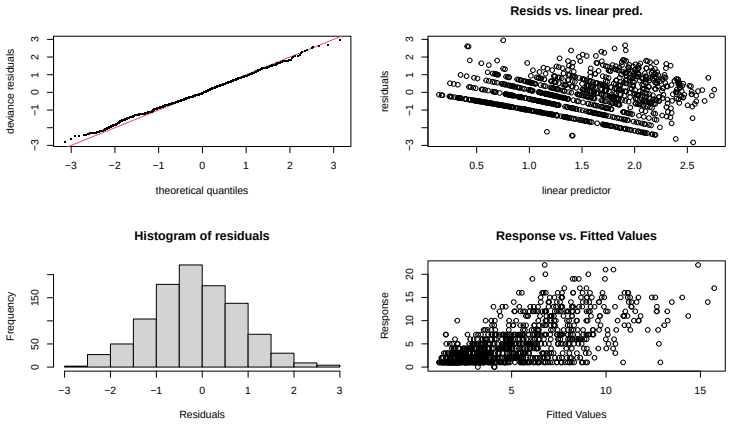


Figure 9: Estimated smoothing curve of the interaction between *catch_agricover* and *catch_sea_accessible*.

4.4 Residual Diagnostics



(a) Connectivity Model Residual Diagnostics



(b) Stressor Model Residual Diagnostics

Figure 10: Residual Diagnostics of the Connectivity and Stressor Model

The residual diagnostics of the stressor and connectivity model can be seen in Figure 10. The QQ-plot of both diagnostics follows a diagonal line with a slight deviance at lower values. The histogram of the residuals follows a symmetric pattern . The residual vs linear predictor in both models is very similar, this may be due to the use of similar typology variables. There is a zero mean in both versions of each plot but there is a clear pattern of horizontal lines in both plots moving from left to right, these lines are due to the fact that the response is discrete and not continuous. The response vs fitted follows a diagonal pattern in both models but it can be seen that the model struggles to fit higher values of *richness* often fitting them less than the true observation.

5 Conclusion

This project examined the relationships between a variety of variables on the number of different fish species counts in UK freshwaters. There were three main research questions in which the effects of certain variables were investigated: the stressor effects, the connectivity effects and how stressor effects are affected by connectivity.

The results indicated that as the percentage of urban land cover increases the partial effect on *richness* decreases. The opposite can be seen for the percentage of agricultural land cover, as it has a positive increasing partial effect on *richness*.

When exploring the interaction between the stressor variables and connectivity variables it was seen that the interactions followed the same trend, with the interaction between the percentage of agricultural land cover and total perimeter of all other lakes in the catchment divided by the catchment area having a positive increasing affect on *richness*. Whilst the opposite could be said for the interaction between the percentage of urban land cover and the total perimeter of all other lakes in the catchment divided by the area. The only significant interaction was the interaction between the percentage of agricultural land cover and if the catchment was sea accessible for fish. If the catchment was sea accessible then there was a decreasing partial effect on *richness*.

The effect of connectivity on *richness* was challenging to model due to the fact that these variables had a high proportion of zeros. In addition, the connectivity effects are interconnected and the effect on richness cannot be examined in isolation or in an additive way. A more complex model with additional variables would be required to truly capture the relationships.

Future work could include exploring the spatial aspect of the data which is yet to properly be accounted for and further investigating the suspected spatial autocorrelation in the data and how it could be modelled. It also may be beneficial to explore new connectivity variables which could be sampled in the future.

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8 Appendix

8.1 interaction model residual diagnostics

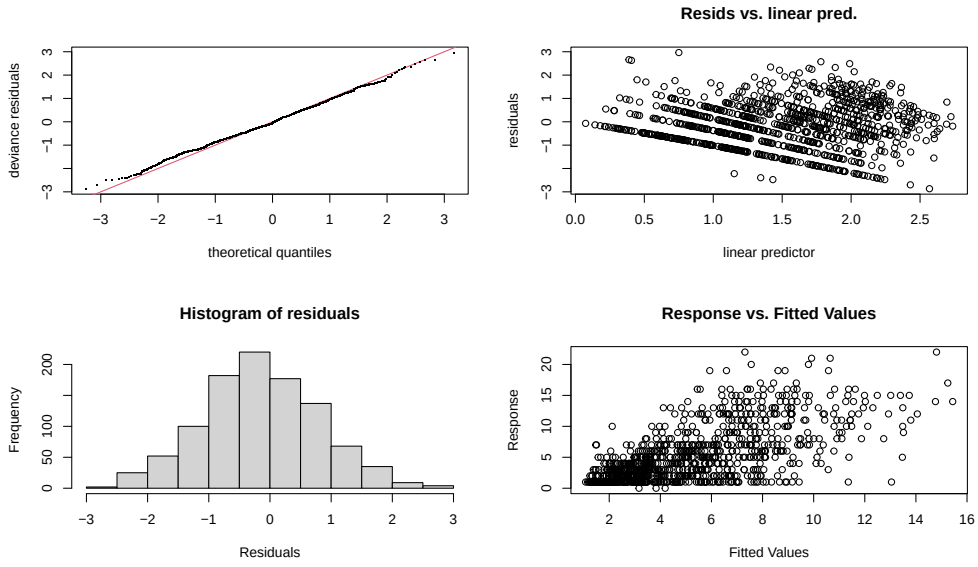


Figure 11: Interaction Model Residual Diagnostics.

8.2 typology variables more examples

An example of more of the estimated smoothing curves. This example was taken from the interaction model.

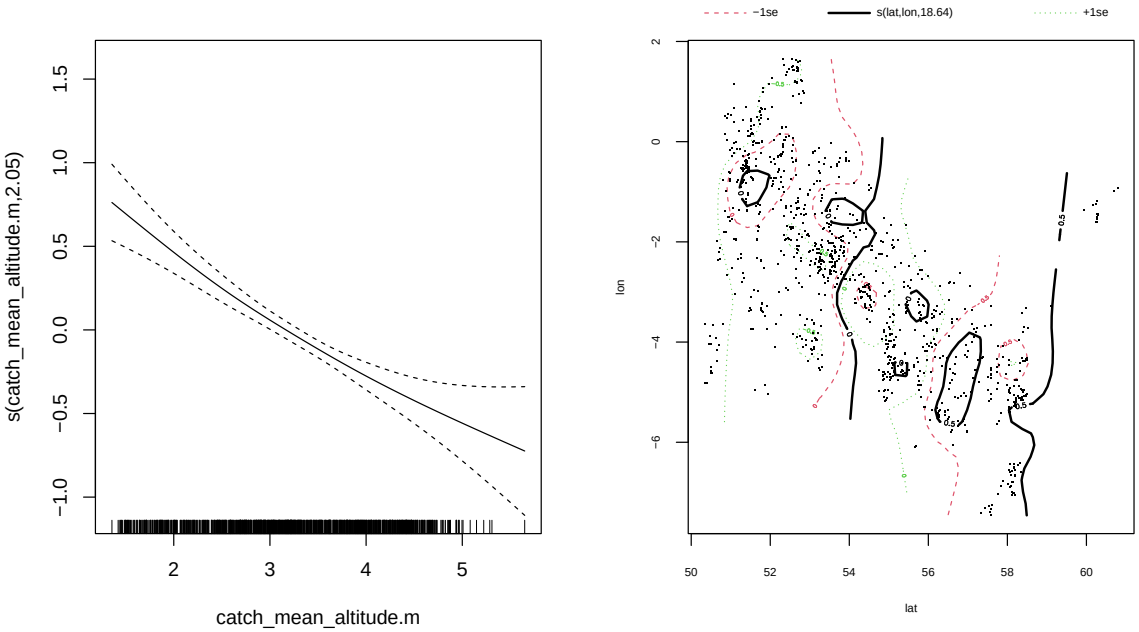


Figure 12: Interaction Model Estimated Smoothing Curves Example.

8.3 data set variables short

Varialble Name (Short)	Variable Name (Full)
lake_area.ha	lake_data_Lake.area..ha.
lake_altitude	lake_data_Lake.altitude..m.
lake_mean_depth.m	lake_data_Mean.Depth..m.
lake_volume.m3	lake_data_Volume..m3.
lake_perimeter.m	lake_data_Perimeter..km.
lake_fetch.km	lake_data_Fetch..km.
lake_straightdistance_tosea.km	lake_data_Distance.to.sea...simple..km.
lake_geotype	lake_data_UK_GEOL_TYPE_
lake_connectivitytype	lake_data_Connectivity.Type
lake_riverbasindistrict	lake_data_WFD_River_Basin_District
catch_mean_altitude.m	catchment_Mean.altitude..m.
mean_slopeoflake_catch	catchment_Mean.slope..degrees.
catch_lakearea.p	catchment_Lake.area..
catch_pondarea.p	catchment_Pond.area..
catch_rivers_length_total	catchment_Rivers...length..m._per_ha
catch_canals_length_total	catchment_Canals...length..m._per_ha
catch_lakes_total_perim	catchment_Lakes...Perimeter..m._per_ha
catch_ponds_total_perim	catchment_Ponds...Perimeter..m._per_ha
catch_lakes_count	catchment_Lakes...Count_per_ha
catch_pond_count	catchment_Ponds...Count_per_ha
catch_visits_fish	catchment_Visits...Fishing
catch_visits_watersport	catchment_Visits...Watersports
catch_visits_all	catchment_Visits...All
catch_obstacles_count	catchment_Obstacles...Count_per_ha
catch_agricover	catchment_LCM2007...Agricultural..
catch_urbancover	catchment_LCM2007...Urban..
catch_sea_accessible	catchment_Salmon_Accessible
catch_mean_temp	catchment_MEAN_mean_temp_2000_2016_pt.tif
catch_mean_annualrainfall	catchment_MEAN_mean_annual_rain_range_2000_2016_pt.tif

8.4 full description data set variables

Variable (Full)	**Description**
lake_data_Lake.area..ha.	Lake area (ha) (t)
lake_data_Lake.altitude..m.	Lake altitude (m above mean sea level) (t)
lake_data_Mean.Depth..m.	Lake mean depth (m) (t)
lake_data_Volume..m3.	Lake volume (m3) (t)
lake_data_Perimeter..km.	Lake perimeter (m) (t)
lake_data_Fetch..km.	Lake fetch (km) (t)
lake_data_Distance.to.sea ...simple..km.	Lake straight-line distance to sea (km) (t)
lake_data_UK_GEOL_TYPE_	Lake geological classification (t)
lake_data_Connectivity.Type	Lake connectivity classification (t)
lake_data_WFD_River_Basin_District	Water Framework Directive River Basin District (t)
catchment_Mean.altitude..m.	Catchment mean altitude (m) (t)
catchment_Mean.slope..degrees.	Mean slope of lake catchment (t)
catchment_Lake.area..	Percent of catchment around lake covered by lakes (c)
catchment_Pond.area..	Percent of catchment around lake covered by ponds (c)
catchment_Rivers...length..m._per_ha	Total length of rivers in catchment divided by catchment area (c)
catchment_Canals...length..m._per_ha	Total length of canals in catchment divided by catchment area (c)

Table 1: Variable descriptions for lake and catchment dataset (Part 1).

Variable (Full)	**Description**
catchment_Lakes...Perimeter ..m._per_ha	Total perimeter of all other lakes in catchment divided by catchment area (c)
catchment_Ponds...Perimeter ..m._per_ha	Total perimeter of all ponds in catchment divided by catchment area (c)
catchment_Lakes...Count_per_ha	Total number of other lakes in catchment divided by catchment area (c)
catchment_Ponds...Count_per_ha	Total number of ponds in catchment divided by catchment area (c)
catchment_Visits...Fishing	Normalised measure of number of visits for fishing purposes (higher = more visits) (c)
catchment_Visits...Watersports	Normalised measure of number of visits for watersports (higher = more visits) (c)
catchment_Visits...All	Normalised measure of number of visits for all purposes (higher = more visits) (c)
catchment_Obstacles...Count_per_ha	Total number of obstacles in catchment divided by catchment area (c)
catchment_LCM2007...Agricultural..	Percent agricultural land cover in lake catchment (s)
catchment_LCM2007...Urban..	Percent urban land cover in lake catchment (s)
catchment_Salmon_Accessible	Indicator of whether lake is accessible to fish from the sea (c)
catchment_MEAN_mean_temp_2000 _2016_pt.tif	Mean temperature of catchment (t)
catchment_MEAN_mean_annual_rain _range_2000_2016_pt.tif	Mean annual rainfall of catchment (t)
lon	Longitude (degrees East) (t)
lat	Latitude (degrees North) (t)

Table 2: Variable descriptions for lake and catchment dataset (Part 2).